Spring 2023 Math 151C Chapter 4 Notes

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Critical Points and Optimization

Extreme Values

Extreme values on an interval are values that are either maximum or minimum values.

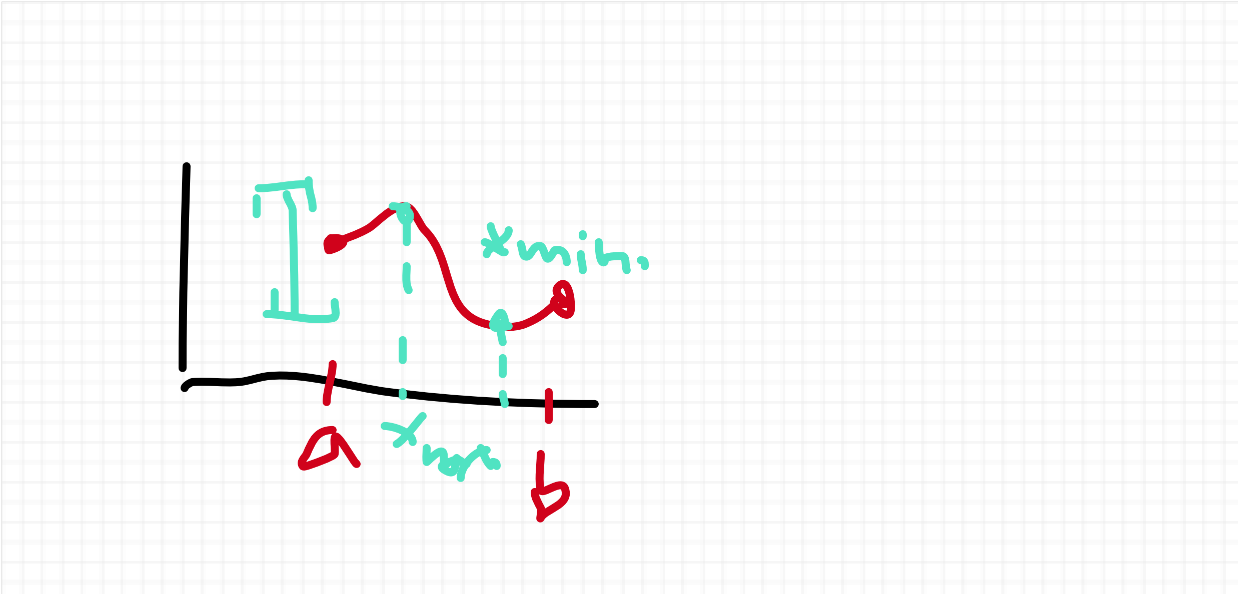
We'll differentiate between so-called local minimum/maximum and absolute minimum/maximum values, but for now, we'll look specifically at absolute minimum/maximum values:

Absolute minimum values on a function f defined on an interval I is the smallest value of the function, i.e. the value $x_{\text{min.}}$ in I such that $f(x_{\text{min.}}) \leq f(x)$, for any other x in I.

Absolute minimum values on a function f defined on an interval I is the largest value of the function, i.e. the value $x_{\text{max.}}$ in I such that $f(x_{\text{max.}}) \geq f(x)$, for any other x in I.

Theorem 1 *(Extreme Value Theorem).* Suppose f is a continuous (by extension this theorem holds for differentiable functions) function defined on a closed bounded interval $[a, b]$. Then there exists both an absolute minimum and absolute maximum.

Proof. We observe--using the figure below--that the values of a continuous function defined on a closed interval always ranges between a closed interval $[a, b]$ of values.

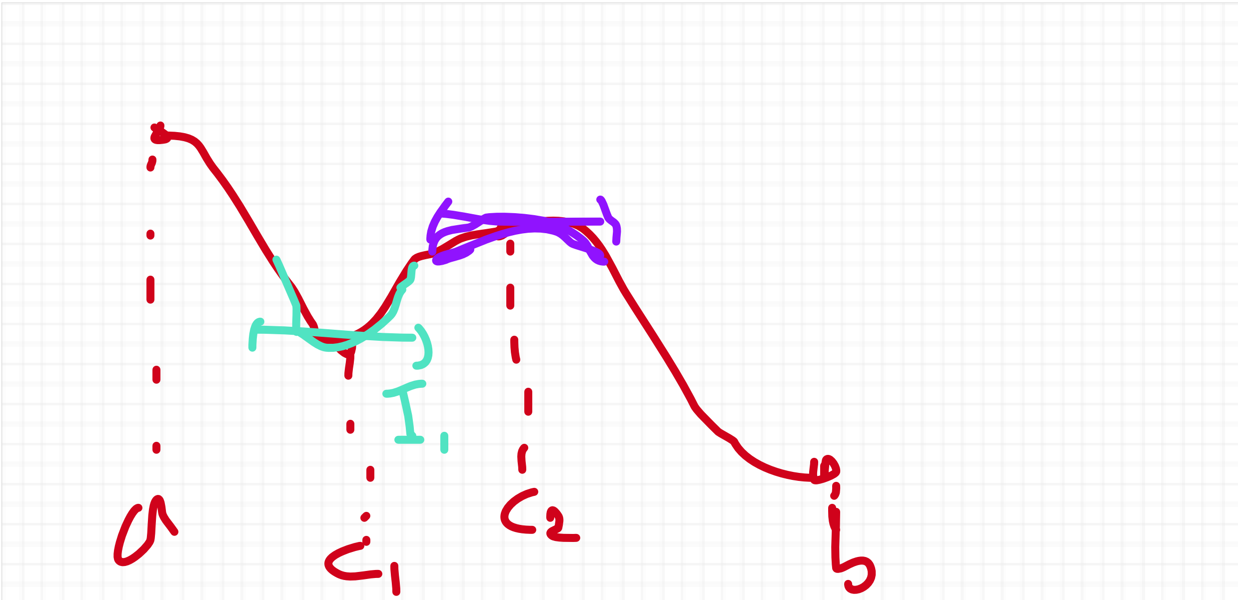


This is formally to say, the image of a continuous function defined on a closed interval is itself a closed interval, giving it a maximum and minimum value. ◻
%TALK MORE ABOUT THE RIGOR BEHIND THE PROOF

Sometimes, a function attains what is called a **local extrema**, which is not itself an absolute extreme value, but becomes one as we zoom closer.

Local minimum. A value c such that $f(c)$ is a minimum in some interval I sufficiently small.

Local maximum. A value c such that $f(c)$ is a maximum in some interval I sufficiently small.



Note that in the diagram above c_1 is a local minimum, since we zoom closer to a smaller interval I_1, the value attained at c_1 is a minimum. Similarly, c_2 is a local maximum, since we zoom closer to a smaller interval I_2, and the value attained at c_2 is a maximum.

Next Time: We'll define what critical points are, and use them to find the extreme values of a function.

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Critical Points

A critical point of a function f is a value of a function c such that $f'(c) = 0$ or is undefined.

NOTE:

1. Usually when dealing with a critical point, we deal with the case where the function is differentiable everywhere and hence whenever a critical point happens, it's because $f'(c) = 0$.

2. The critical point is oftentimes not unique, and the way to find the critical in practice is to find the deritivative function f' of f and then solve for x in the equation $f'(x) = 0$

Example 1 *(from Example 1 on page 188)* Find the critical points of

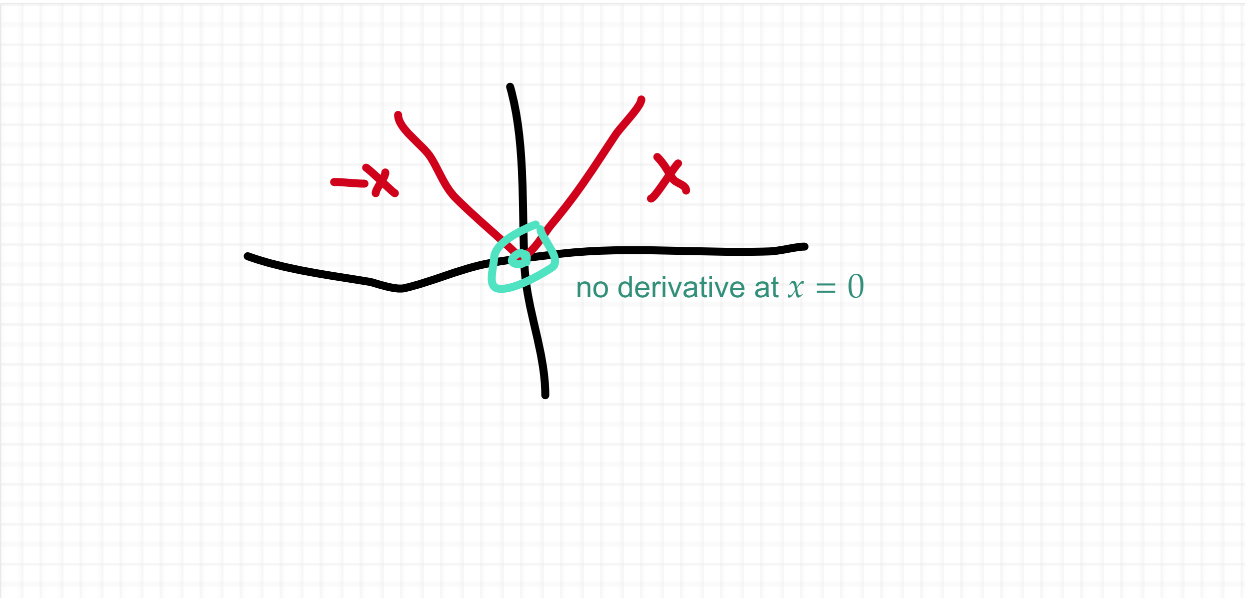
$f(x) = x^3 - 9x^2 + 24x - 10$

$3x^2 - 18x + 24 = f'(x) = 0$

$3\left(x^2 - 6x + 8\right) = 0$
$3(x-4)(x-2) = 0$

We see that x has the solutions $4, 2$ in the equation, so the critical points are $x = 4, 2$

Example 2. *(from Example 2 on page 188)* Find the critical points of $f(x) = |x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$. If we recall this example from chapter 3. The derivative is as follows:



$$f'(x) = \begin{cases} 1 & \text{if } x > 0 \\ -1 & \text{if } x < 0 \\ DNE & \text{if } x = 0 \end{cases}$$

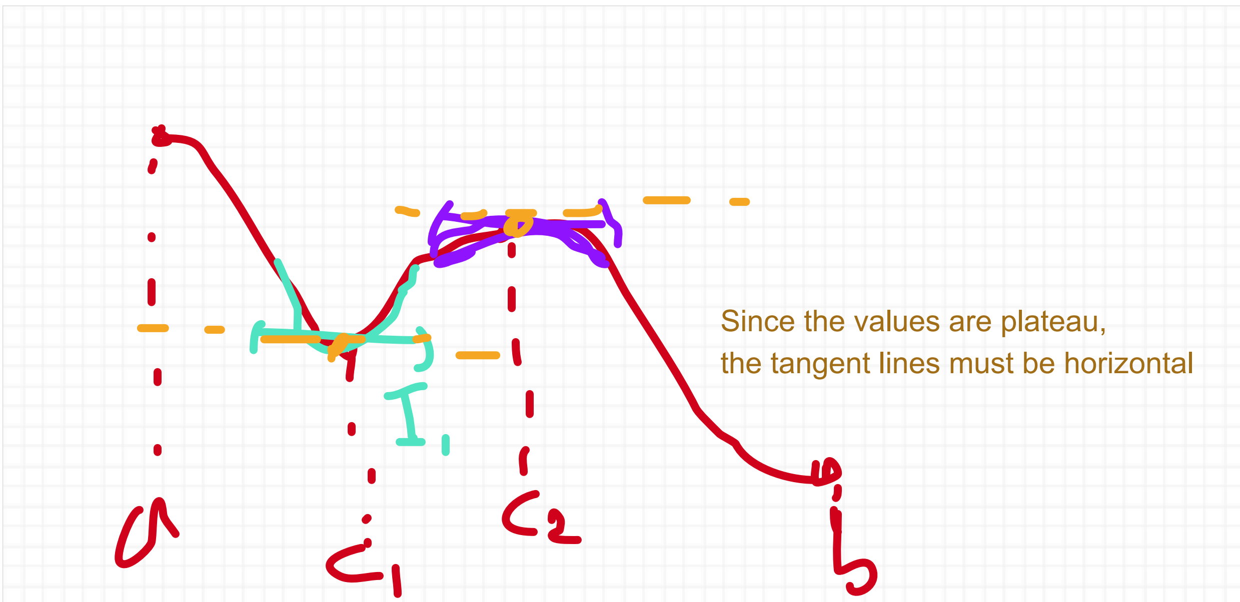
The single critcal point is $x = 0$ where it's undefined, and nowhere else since the derivative is nonzero elsewhere.

Optimization on a Closed Interval

Now that we know how to find a critcial point, we are in a position to optimize the a function on a closed interval.

Theorem 2. *(Fermat's Theorem on Loal Extrema)* If $f(c)$ is a local minimum or maximum, then c is a critical point of f.

Proof.



□

%TALK MORE ABOUT THE RIGOR BEHIND THE PROOF

How do we find extrema? To answer this question, note that an absolute minimum (resp. maximum) must either be a local minimum (resp. absolute minimum), or an endpoint, since any non-endpoint minimum (resp. maximum) must be also be a local minimum (resp. maximum), hence Fermat's thoerem tells us in the case where $f(x)$ is a non-endpoint minimum (resp. maximum) is the absolute minimum (resp. maximum).

This gives us the following theorem that specifies the possible candidates for the minimum and maximum.

Theorem 3. An absolute minimum (resp. maximum) of a function f defined on a closed interval $[a, b]$ is either a critical point, or an endpoint.

This gives us the following step-by-step process on how to find the extreme values (i.e. the absolute minimum and maximum):

Step 1. Find $f'(x)$ and set it equal to 0, so that we have $f'(x) = 0$.

Step 2. Solve for x to find all the critical points $c_1, \ \dots, c_n$ in $[a, b]$

Step 3. Plug in all the the values of the critical points and endpoints $f(a), \ f(c_1), \ \dots, f(c_n), \ f(b)$. The lowest value is the absolute minimum, and the highest value is the absolute maximum.

Example 3. Find the extrema of $f(x) = 2x^3 - 15x^2 + 24x + 7$ on $[0, 6]$.

Step 1. We find that

$$f'(x) = 6x^2 - 30x + 24$$

and so we have the equation

$$6x^2 - 30x + 24 = f'(x) = 0$$

Step 2. We solve for x for <u>critical points in the interval $[0, 6]$.</u>

$$6\left(x^2 - 5x + 4\right) = 0$$
$$6(x - 4)(x - 1) = 0$$

so we have the solutions $x = 4, 1$, which are critical points in the interval $[0, 6]$

Step 3. We want to plug in all the critical points plus endpoints as so:

$$f(0) = 7,$$
$$f(1) = 2 - 15 + 24 + 7 = 18$$
$$f(4) = -9$$
$$f(6) = 43$$

So the lowest and highest values, respectively, are $f(4) = -9$ and $f(6) = 43$, so these are the absolute minimum and maximum of $f(x)$ defined on $[0, 6]$.

Example 4. *(from Example 5 on page 190)* Find the extreme values of $f(x) = \dfrac{x}{x^2 + 5}$ on $[4, 8]$

$$f'(x) = \frac{(1)\left(x^2+5\right) - x(2x)}{\left(x^2+5\right)^2} = \frac{-x^2+5}{\left(x^2+5\right)^2} = 0$$

We solve for the equation above as follows:

Note that $\left(x^2+5\right)^2 \neq 0$ for all x on the interval $[4, 8]$, so then we can multiply that term away since it is nonzero

$$\frac{-x^2+5}{\left(x^2+5\right)^2} = 0$$
$$\cdot - \left(x^2+5\right)^2 \quad \cdot - \left(x^2+5\right)^2$$
$$x^2 - 5 = 0$$
$$\left(x+\sqrt{5}\right)\left(x-\sqrt{5}\right) = 0,$$

so the critical points are $x = \pm\sqrt{5} \approx \pm 2.24$, but both of these points are outside the interval $[4, 8]$, so to find the absolute minimum and maximum, we only plug in the endpoints

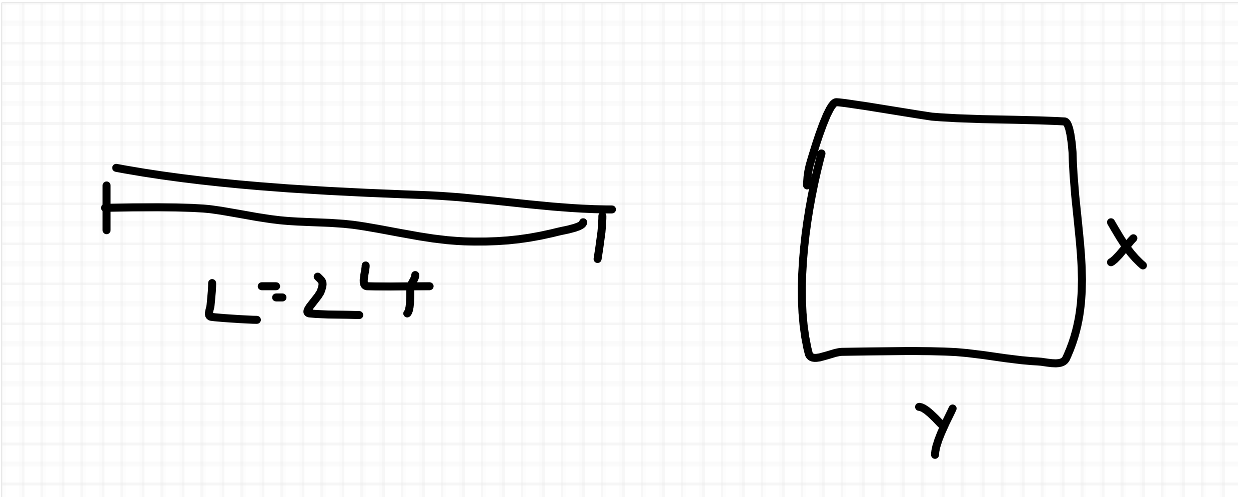
$$f(4) = \frac{4}{21},$$
$$f(8) = \frac{8}{69},$$

so the absolute minimum is $f(8) = 8/69$, and the absolute maximum is $f(4) = 4/21$.

Applied Optimization

Here, we want to optimize a typically two variable problem with a constraint, which will allow us to use what we learned about optimizing on a closed interval to optimize the applied problem

Example 1. A piece of wire of length $L = 24$ is bent into the shape of a rectangle. Which dimensions produce the rectangle of maximum area



Note that the total length is the perimeter of the rectangle, which is $24 = 2x + 2y$, so then we can express y in terms of x as so:

$$y = \frac{24 - 2x}{2} = 12 - x$$

so the area A can be given as a function of x, defined by

$$A(x) = xy = x(12 - x) = 12x - x^2$$

and the least that x can be is 0 and the most is 12, so then optimizing the area comes down to optimizing $A(x)$ on the interval $[0, 12]$, so we find the critical point, which is

$$\begin{aligned} A'(x) &= 0 \\ 12 - 2x &= 0 \\ x &= 6 \end{aligned}$$

So then we plug in the critical point with the endpoints and find that

$$\begin{aligned} A(0) &= 0(12 - 0) = 0 \\ A(6) &= 6 \cdot 6 = 36 \\ A(12) &= 12(12 - 12) = 0 \end{aligned}$$

so as a result, the maximum is $A(6) = 36$.

Next Time: We'll give the general step-by-step process on how to set up such problems, do another example, and move on to the rest of Chapter 4.

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Optimization problems usually have an equation (usually two variables) that we're optimizing and a constraint between the two variables. The constraint allows to set one variable in terms of another, and what we end up with is optimizing over an interval. This gives us the following three general steps:

Step 1. Identify the variables.
Step 2. Convert the two variable function to a single variable function using the constraint.
Step 3. Optimize using the opimization technique for a function defined on an interval (using the three step process for that)

Example 2. *(from Example 2 on page 219-220)* Our task is to minimize the travel time in building a road that connects a small town from a capital city. If the speed limit is 60km/hour on the road and 110km/hour on route 1. The perpendicular distance from the small town to route 1 is 30km vertically and 50km horizontally.

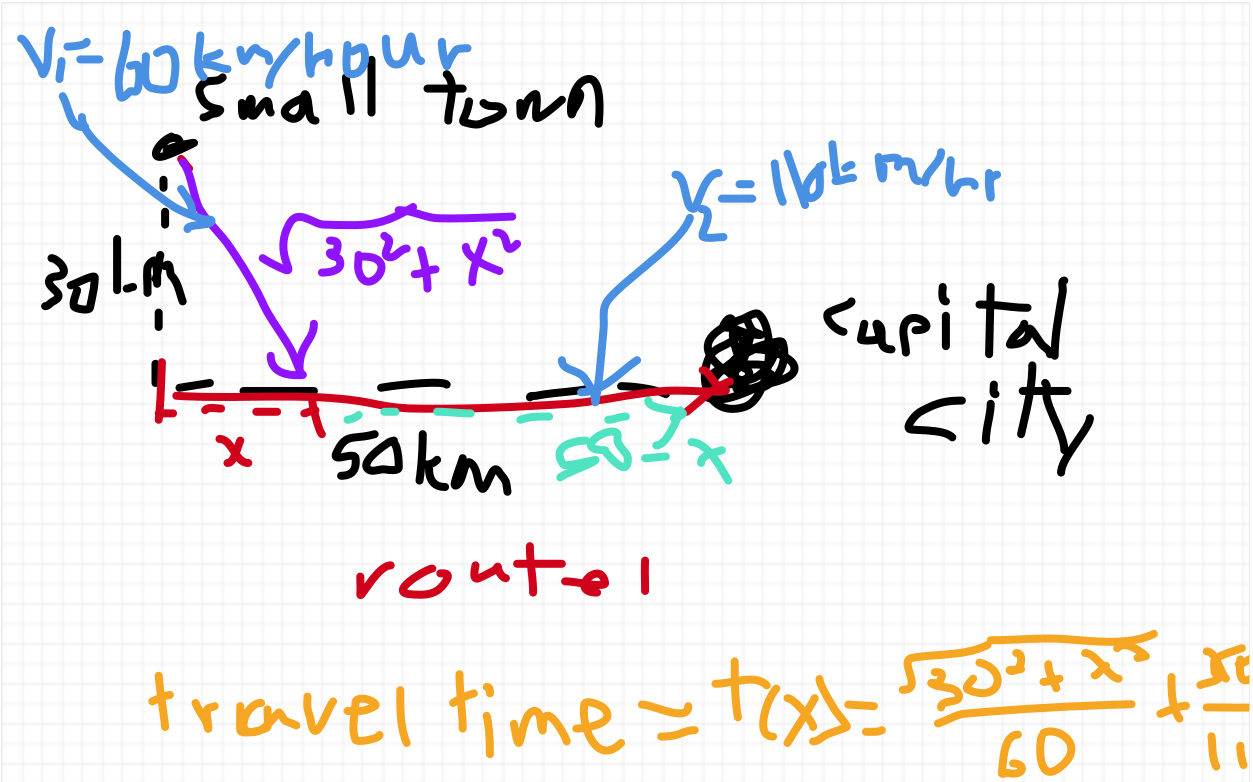
Step 1. In doing the first step, it often helps to draw a picture of the problem itself (since often making sense of the variables and converting them involves geometry).

Recall that the travel time can be determined by distance / velocity, since distance d for velocity v that is linear is determined by

$d = vt,$
$t = d/v$

The time traveled is equal to the time t_1 traveled on the road that we're building (in purple) plus the time t_2 traveled on route 1 (in red), so we have

$t_1 = d_1/v_1,\ t_2 = d_2/v_2$
$v_1 = 60,\ d_1 = \sqrt{30^2 + x^2},\ v_2 = 110,\ d_2 = 50 - x$



Step 2. Note that $0 \le x \le 50$

We determine that we are minimizing the function

$t(x) = \dfrac{\sqrt{30^2 + x^2}}{60} + \dfrac{50 - x}{110}$ over the interval $[0, 50]$

And so now we are ready to do Step 3.

Step 3. We know that the end points are $x = 0, 50$, and then we find the critical points by setting

$t'(x) = 0$

$$t'(x) = \frac{d}{dx}\left(\frac{\sqrt{30^2 + x^2}}{60}\right) + \frac{d}{dx}\left(\frac{50 - x}{110}\right)$$
$$= \frac{1}{60} \cdot \frac{1}{2}\left(30^2 + x^2\right)^{-1/2} - \frac{1}{110}$$

So now we colve for x with the equation

$$
\begin{aligned}
\frac{1}{60}\cdot\frac{1}{2}\left(30^2+x^2\right)^{-1/2}2x-\frac{1}{110} &= 0\\
\frac{1}{60}\left(30^2+x^2\right)^{-1/2}x &= \frac{1}{110}\\
\times 110\left(900+x^2\right)^{1/2}x \qquad & \times 110\left(900+x^2\right)^{1/2}\\
\frac{11}{6}x &= \left(900+x^2\right)^{1/2}\\
(\ )^2 \qquad & (\ )^2\\
\frac{121}{36}x^2 &= 900+x^2\\
-x^2 \qquad & -x^2\\
\frac{85}{36}x^2 &= 900\\
(\ )^{1/2} \qquad & (\ )^{1/2}\\
\frac{\sqrt{85}}{6}x &= 30\\
x &= \frac{180}{\sqrt{85}}\approx 19.52
\end{aligned}
$$

The candidates for the minimum are $0, 19.52, 50$. If we plug them in, we get

$$t(0) = 0.95,\ t(19.52)\approx 0.87,\ t(50) = 0.97,$$

between all that, we determine that the critical point (which we worked stinking hard to get!) is the minimum! (so all that hard work didn't go in vein).

Behavior of a Function

Mean Value Theorem

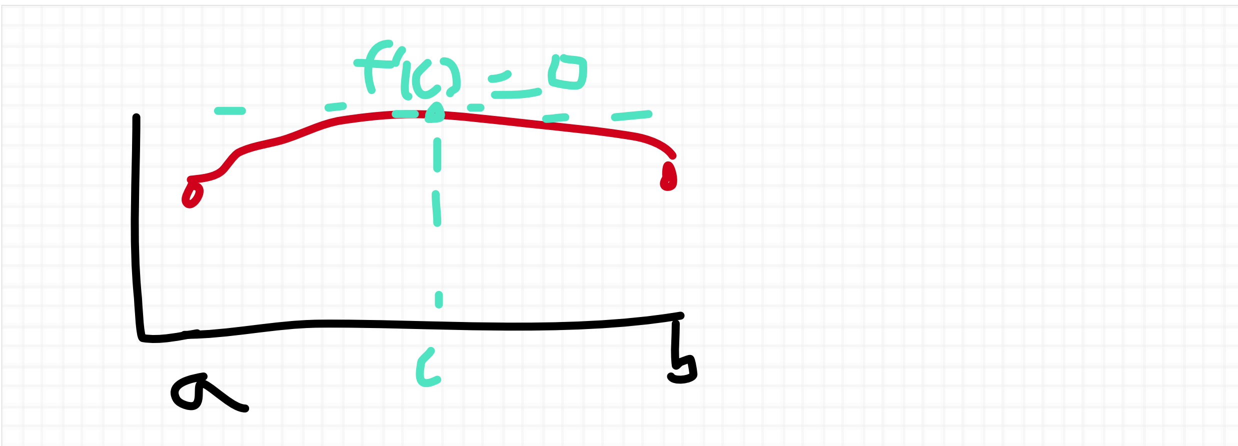
Theorem 1. *(Rolle's Theorem)* Assume that f is continuous on $[a, b]$ and differentiable on (a, b). If $f(a) = f(b)$, then there exists a number between a and b such that $f'(c) = 0$

Proof. Note in general (by the **Extreme Value Theorem**) that the function has a minimum and maximum value. If the minimum and maximum values agree with the endpoints, i.e.

$f(x_{min}) = f(x_{max}) = f(a) = f(b)$

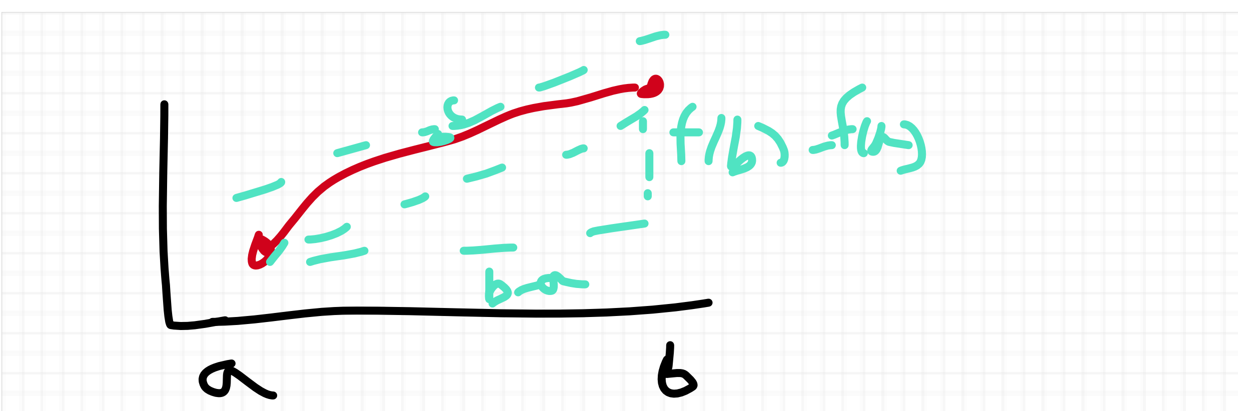
then the function on the interval is constant, and hence any value in (a, b) has a zero derivative.

Otherwise, we have an absolute extrema (either a minimum or maximum) $f(c) \neq f(a), f(b)$ for some $a < c < b$ that is also a local extrema (either a local minimum or local maximum).



Our conclusion in this case that $f'(c) = 0$ follows by **Fermat's Theorem.** $\square$

Theorem 2. *(Mean Value Theorem)* Assume that f is continuous on $[a, b]$ and differentiable on (a, b). Then there exists a number c between a and b such that $f'(c)$ is the "mean value" $\dfrac{f(b) - f(a)}{b - a}$, i.e. we have $f'(c) = \dfrac{f(b) - f(a)}{b - a}$



Proof. Follows immediately by Rolle's theorem applied to the function

$$G(x) = f(x) - \frac{f(b) - f(a)}{b - a}(x - a),$$

that there exists $a < c < b$ such that $G'(c) = 0$, hence

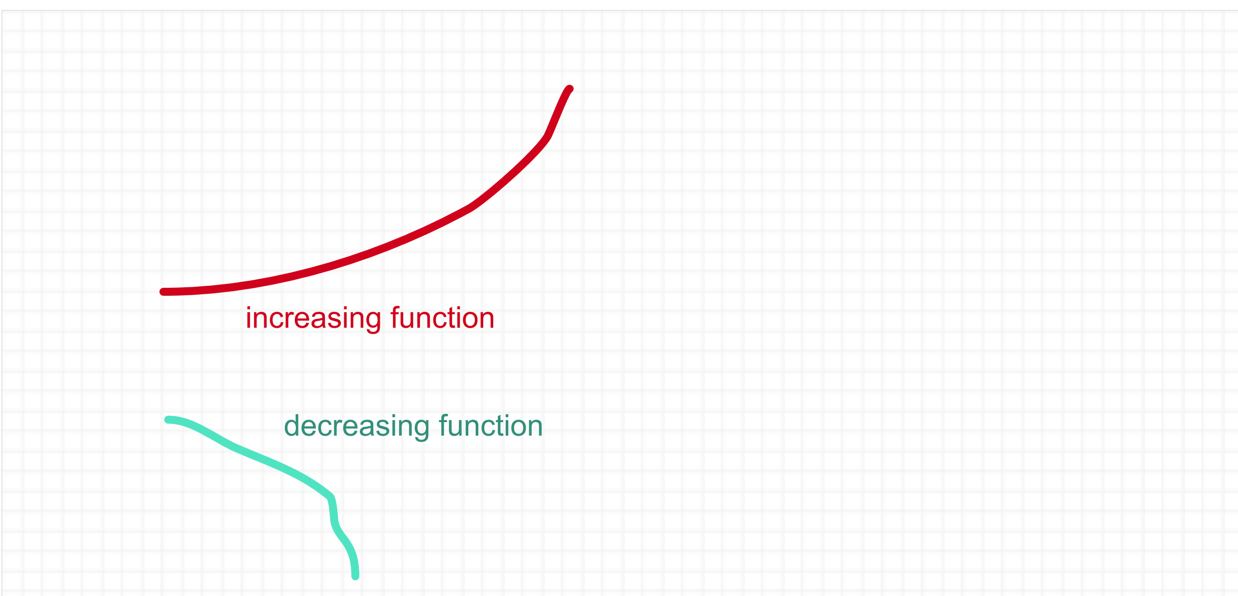
$$0 = \frac{d}{dx}[f(x)] - \frac{d}{dx}\left[\frac{f(b) - f(a)}{b - a}(x - a)\right]\Bigg|_{x=c}$$
$$= f'(c) - \frac{f(b) - f(a)}{b - a},$$

$$\implies f'(c) = \frac{f(b) - f(a)}{b - a},$$

and our conclusion is met. □

Increasing/Decreasing Conditions on Functions

Note that f is **(strictly) increasing** if $x < y \implies f(x) < f(y)$, and f is **(strictly) decreasing** if $x < y \implies f(x) > f(y)$



Theorem 3. Let f be a differentiable function on (a, b)

1. $f'(x) > 0$ for all x in (a, b) implies f is increasing.
2. $f'(x) < 0$ for all x in (a, b) implies f is decreasing.

Proof. Note that if f is not increasing then there is some $x < y$ in (a, b) such that $f(x) > f(y)$, and by the **Mean Value Theorem** we have some $a < x < c < y < b$ such that

$$f'(c) = \frac{f(y) - f(x)}{y - x} < 0,$$

contradicting our hypothesis that $f'(x) > 0$ for all x in (a, b). We proceed with the proof similarly with showing $f'(x) < 0$ for all x in (a, b) implies f is decreasing. $\square$

Example 1. *(from Example 2 on page 198)* Show that the function $f(x) = 3x - \cos 2x$ is (always) increasing.

Solution. Find the derivative of $f(x)$ as so

$$f'(x) = 3 + 2\sin 2x$$

and $-2 \leq 2\sin 2x \leq 2$, therefore $f'(x) > 0$ always, and we conclude it's increasing.

Next Time: We'll do another example of finding monotonic values, and then do the first and second derivative tests.

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To find whether a function f is increasing or decreasing on a differentiable function, it suffices to find the intervals where $f'(x) > 0$ (which implies that it's increasing) and where $f'(x) < 0$ (which implies that it's decreasing). We do this in the following steps:

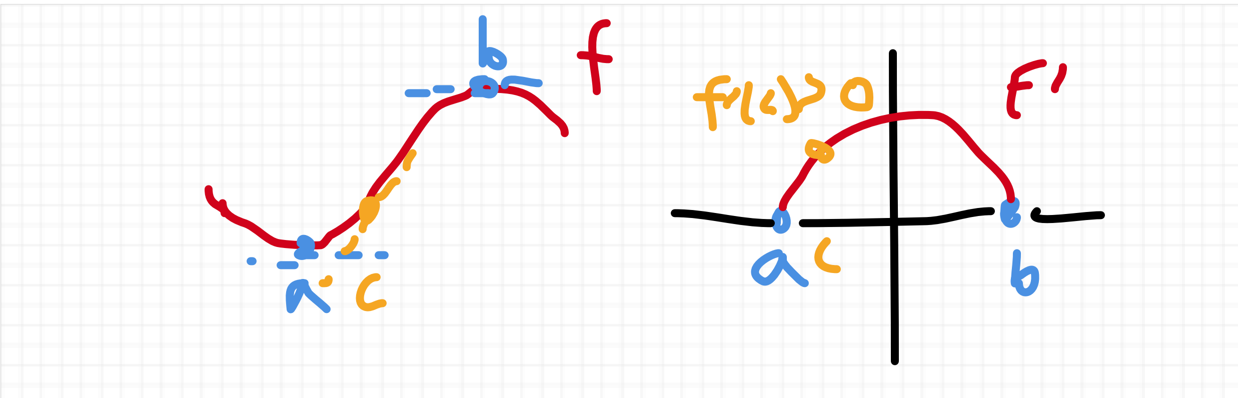
Step 1: Find the derivative of f.

Step 2: Find the critical points of f

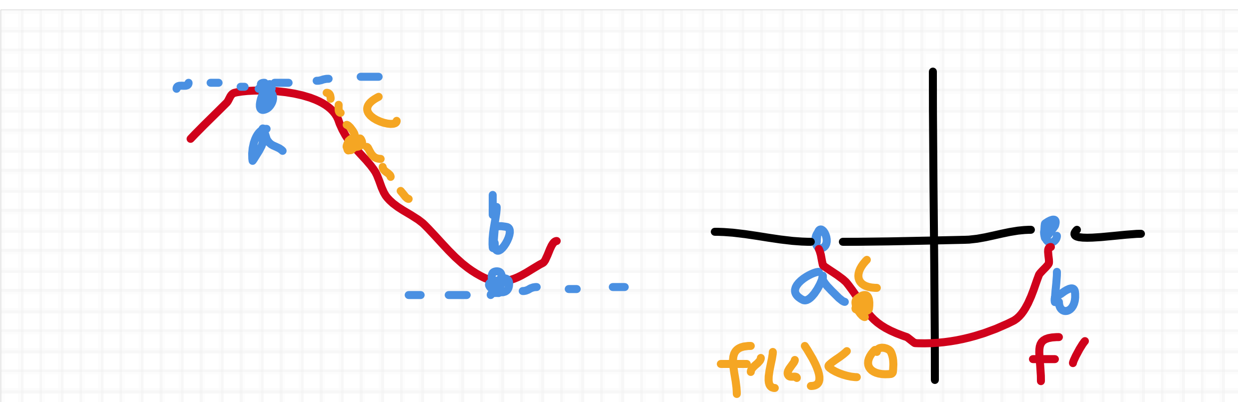
Step 3: Segment the intervals between the critical points and figure whether those intervals are negative ($f'(x) < 0$, i.e. f increasing) and positive ($f'(x) > 0$, i.e. f is decreasing); it suffices to plug in a single point to figure out whether the derivative is negative or positive (or we of course can go by vibes). We shall declare this fact with the following theorem:

Theorem 4. Let f be a differentiable function on $[a,b]$ such tha f' is continuous on $[a,b]$, $x = a, b$ are critical points, and there are no other critical points in (a,b).

1. $f'(c) > 0$ for *some* c in (a,b) implies $f'(x) > 0$ *for all* x in (a,b) (which in turn implies f is increasing.on (a,b))



2. $f'(c) < 0$ for *some* c in (a,b) implies $f'(x) > 0$ *for all* x in (a,b) (which in turn implies f is decreasing.on (a,b))



Proof. We find that in the hypothesized scenario where $f'(c) > 0$ (resp. $f'(x) < 0$) for some c in (a,b), it must hold that $f'(x) > 0$ (resp. $f'(x) < 0$) for all x in (a,b), or else in the case where $f'(x) < 0$ (resp. $f'(x) > 0$) for any other x in (a,b), we find since f' is continuous on (a,b) by the **Intermediate Value Thoerem** that there exists x' between x and c such that $f'(x') = 0$, contradicting the fact that there are no critical points in (a,b) $\square$

NOTE: The conclusion of **Theorem 4** can actually be proven without assuming continuity of

f' using what is called **Darboux's Thoerem**, or The **Intermediate Value Thoerem for Derivatives** in place of the regular **Intermediate Value Theorem**. Going into detail on this, however, goes far beyond the scope of this course.

Example 2. *(from Example 3 on page 198)* Find the intervals on which $f(x) = x^2 - 2x - 3$ is increasing/decreasing.

Step 1. we want to find the derivative

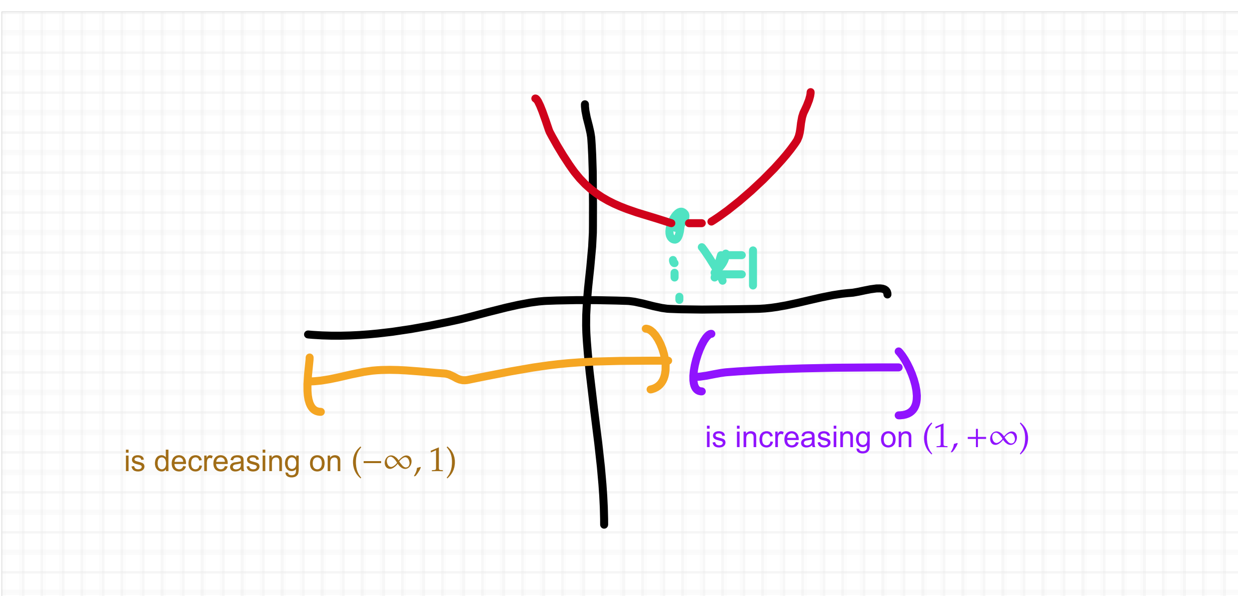
$$f'(x) = 2x - 2$$

Step 2. Find the critical point(s) of f

$$0 = f'(x) = 2x - 2 \implies x = 1$$

Step 3. We have the intervals $(-\infty, 1)$ and $(1, +\infty)$ where $f'(x) \neq 0$, so we can then figure out which intervals are positive and which ones are negative. We can see that $f'(x) = 2x - 2$ is negative at $(-\infty, 1)$ and positive at $(1, +\infty)$ and we can verify that by plugging in 0 and 2 respectively,

$f'(0) = -2 \implies f$ is decreasing on $(-\infty, 1)$
$f'(2) = 2 \implies f$ is increasing on $(1, +\infty)$



Example 3. Find the critical points and intervals where the function is increasing and decreasing for $f(x) = 5x^2 + 6x - 4$

We find

$$f'(x) = 10x + 6$$

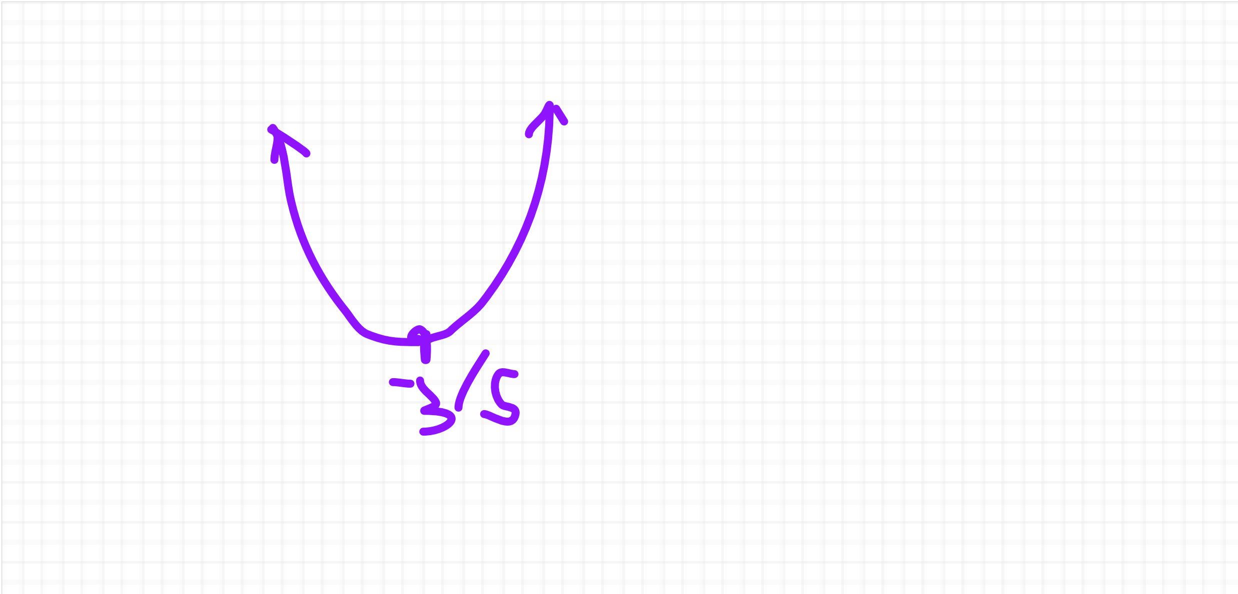
To find the critical point, we set f' equal to 0 and solve for x:

$$\begin{aligned} 10x + 6 &= 0 \\ -6 \quad & \quad -6 \\ 10x \quad &= -6 \\ \div 10 \quad & \quad \div 10 \\ x \quad &= \frac{-6}{10} = -\frac{3}{5} \end{aligned}$$

Next, we find behavior of f over the intervals $\left(-\infty, -\frac{3}{5}\right)$ and $\left(-\frac{3}{5}, +\infty\right)$ between the critical point $-\frac{3}{5}$. Plugging in -1 and 0 in each interval respectively, we get

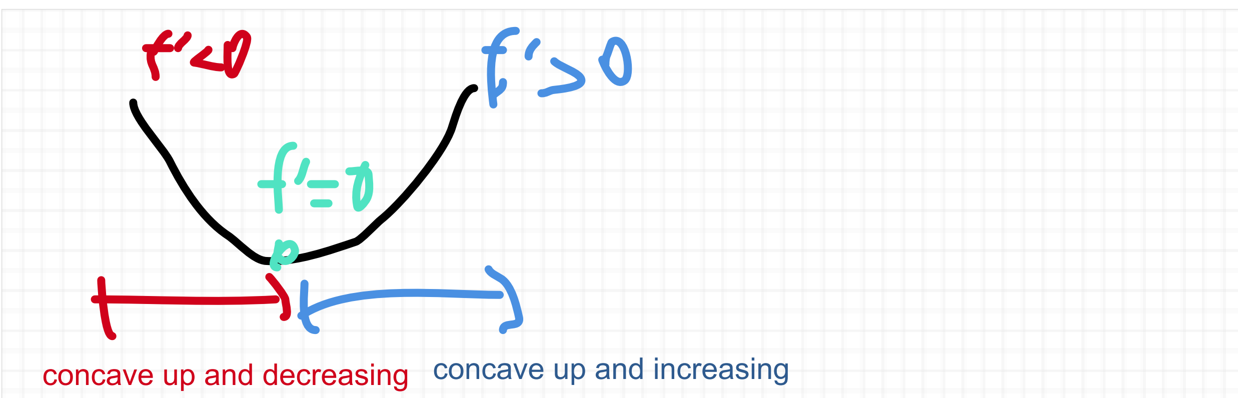
$f'(-1) = -4 \Longrightarrow f$ is decreasing on $\left(-\infty, -\frac{3}{5}\right)$

$f'(0) = 6 \Longrightarrow f$ is increasing on $\left(-\infty, -\frac{3}{5}\right)$



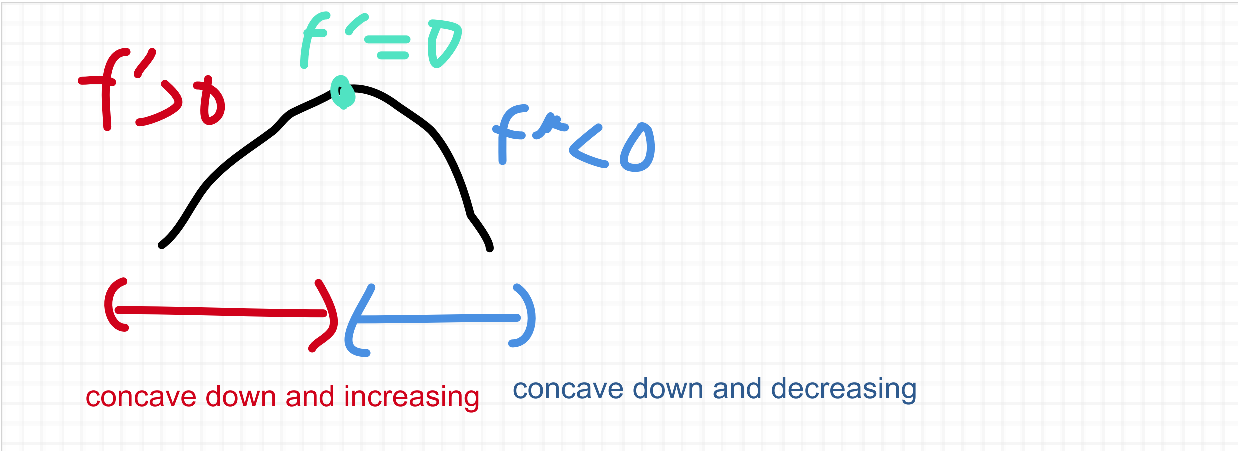
Concavity and Inflection Points

A function is **concave up** (or **convex**) if f' is increasing.



Notice in the diagram above that f is concave up precisely when f is either decreasing progressively slower (see the red part of the curve above) or increasing progressively faster (see the blue part of the curve above). In both instances, you have f' increasing

A function is **concave down** (or **concave**) if f' is decreasing.



Notice in the diagram above that f is concave down precisely when f is either increasing progressively slower (see the red part of the curve above) or decreasing progressively faster (see the blue part of the curve above). In both instances, you have f' decreasing.

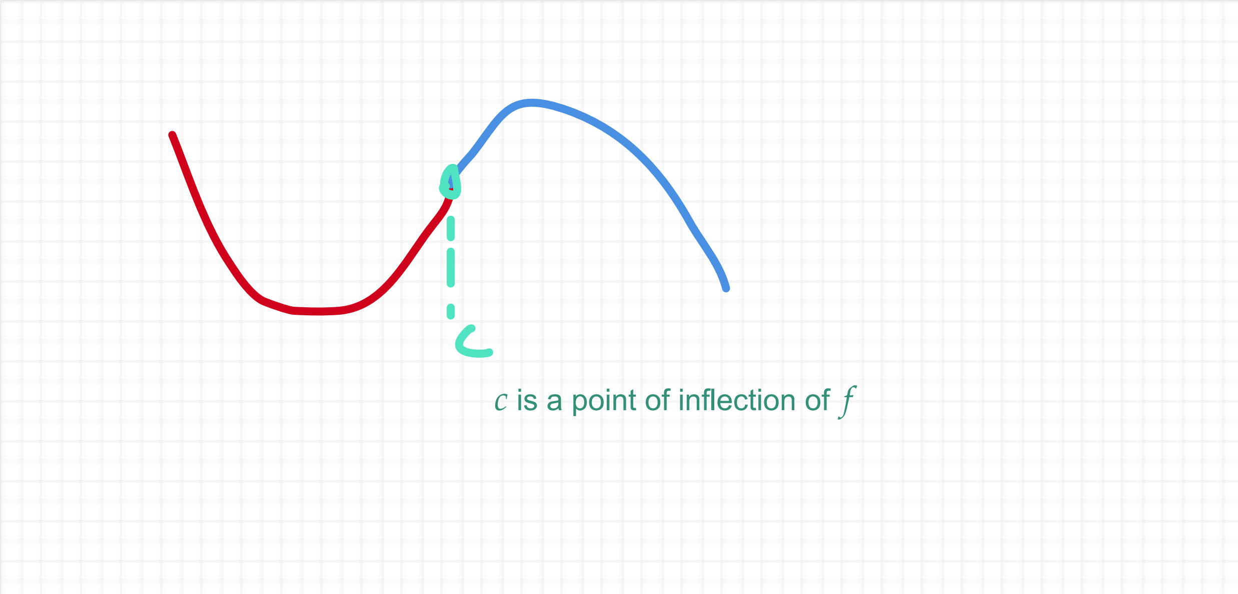
Concavity corresponds to whether the shape of the function if so-called "concave" or "convex" (like in physics), but how the shape is done can explained using its first or second derivative.

Theorem 1. Let f be a twice differentiable function on (a, b)

1. f is concave up on (a, b) if $f''(x) > 0$.
2. f is concave down on (a, b) if $f''(x) < 0$.

Proof. Follows immediately from the fact that $f'(x) > 0 \implies f$ is increasing and $f''(x) < 0 \implies f$ is decreasing. $\square$

A **point of inflection** is a point where it changes concavity,



Theorem 2.

1. If f is twice differentiable, and at point c $f''(c) = 0$, and f'' changes signs from the right and the left of c, then c is a point of inflection of f.

2. If c is a point of inflection and f is twice differentiable and f'' is continuous, then $f''(c) = 0$ and c is a critical point of f

Proof.

1. Follows immediately by **Theorem 1**, since $f''(x) > 0 \implies f$ is concave up and $f''(x) < 0 \implies f$ is concave down.

2. Since f'' is continuous, the intermediate value thoerem tells us that there must exist some a as $x \to c$ such that $f''(a) = 0$ since we have $f''(x) > 0$ and $f''(x) < 0$. We find by default this point is c, since $f''(x) \neq 0$ everywhere else. $\square$

Our takeaway from **Theorem 2** is that we find the points of inflection simply by finding where $f''(x) = 0$ and f'' changes sign. Since we finding concavity is by definition where f' is increasing/decreasing, we can do this using the same method three step method as before, except for f', i.e.:

Step 1: Find f''.

Step 2: Find the critical points of f' (i.e. where $f''(x) = 0$)

Step 3: Segment the intervals between the critical points and figure whether those intervals are negative ($f''(x) < 0$, i.e. f is concave up) and positive ($f''(x) > 0$, i.e. f is concave down); it suffices to plug in a single point to figure out whether the second derivative is negative or positive.

Example 1 *(from Example 3 on page 205)*. Find the points of inflection and concavity on each interval between the points of inflection the function $f(x) = 3x^5 - 5x^4 + 1$

$$f'(x) = 15x^4 - 20x^3 = 5\left(3x^4 - 4x^3\right) = 5(3x - 4)x^3$$
$$f''(x) = 60x^3 - 60x^2 = 60x\left(x^2 - 1\right) = 60(x + 1)x(x - 1)$$

The equation $f''(x) = 0$ has the solution $x = -1, 0, 1$, which are all possible points of inflection

Finding concavity is the same as finding whether f' is increasing or decreasing, so to do that, we plug in the points on $f'(x)$ between the intervals $(-\infty, -1), (-1, 0), (0, 1), (1, +\infty)$. We shall pick $-2, -1/2, 1/2, 2$

$f''(-2) < 0 \Longrightarrow f'$ is decreasing $\Longrightarrow$ concave down
$f''(-1/2) > 0 \Longrightarrow f'$ is increasing $\Longrightarrow$ concave up
$f''(1/2) < 0 \Longrightarrow f'$ is decreasing $\Longrightarrow$ concave down
$f''(2) > 0 \Longrightarrow f'$ is increasing $\Longrightarrow$ concave up

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Example 2. *(from 4.4 exercise 17 on page 208)* Find the intervals of concavity and the points of inflection for the function

$$f(x) = \frac{x^3}{1 + x}$$

First, we want to find f'', and we'll do so as follows

$$f'(x) = \frac{3x^2(1 + x) - x^3(1)}{(1 + x)^2} = \frac{3x^2 + 3x^3 - x^3}{(1 + x)^2} = \frac{3x^2 + 2x^3}{(1 + x)^2}$$

$$f''(x) = \frac{d}{dx}[f'(x)] = \frac{\frac{d}{dx}\left[3x^2 + 2x^3\right](1+x)^2 - \left(3x^2 + 2x^3\right)\frac{d}{dx}\left[(1+x)^2\right]}{\left((1+x)^2\right)^2}$$

$$= \frac{\left(6x + 6x^2\right)(1+x)^2 - \left(3x^2 + 2x^3\right)2(1+x)(1)}{(1+x)^4}$$

$$= \frac{\left[\left(6x + 6x^2\right)(1+x) - 2\left(3x^2 + 2x^3\right)\right](1+x)}{(1+x)^4}$$

$$= \frac{\left[6x + 6x^2 + 6x^2 + 6x^3\right] - \left[6x^2 + 4x^3\right]}{(1+x)^3}$$

$$= \frac{6x + 6x^2 + 2x^3}{(1+x)^3}$$

$$= \frac{2x\left(x^2 + 3x + 3\right)}{(1+x)^3}$$

For this exercise (and similarly exercise 16 on your homework), we want to find the inflection points by finding the solution to $f''(x) = 0$ and we also want to find where f'' is undefined by finding the roots of the numerator, which in this case is simply $x = 0$, and also note that at $x = -1$, it's undefined

$$0 = f''(x) = \frac{2x\left(x^2 + 3x + 3\right)}{(1+x)^3}$$

$$\div \frac{2\left(x^2 + 3x + 3\right)}{(1+x)^3} \quad \div \frac{2\left(x^2 + 3x + 3\right)}{(1+x)^3}$$

$$0 \qquad = x$$

So now we look at the intervals $(-\infty, -1), (-1, 0),\ (0, +\infty)$ between the zeros of the function, and plug in a single value and find whether the second derivative is positive or negative. Let's choose $x = -2$ for $(-\infty, -1)$, $x = -1/2$ for $(-1, 0)$ and $x = 1$ for $(0, +\infty)$

$$f''(-2) = \frac{2(-2)\left((-2)^2 + 3(-2) + 3\right)}{(1 + (-2))^3} = \frac{-4(4 - 6 + 3)}{(-1)^3} = \frac{-4(1)}{-1} = 4 > 0$$

$\implies f''(x)$ is positive on $(-\infty, -1)$

$$f''(-1/2) = \frac{2(-1/2)\big((-1/2)^2 + 3(-1/2) + 3\big)}{(1 + (-1/2))^3} = \frac{-1(1/4 - 6/4 + 12/4)}{(1/2)^3} = \frac{-1(7/4)}{(1/2)^3} < 0$$

$\implies f''(x)$ is negative on $(-1, 0)$

$$f''(1) = \frac{2(1)\big((1)^2 + 3(1) + 3\big)}{(1 + (1))^3} = \frac{2(7)}{2^3} = \frac{7}{4} > 0$$

$\implies f''(x)$ is positive on $(0, +\infty)$

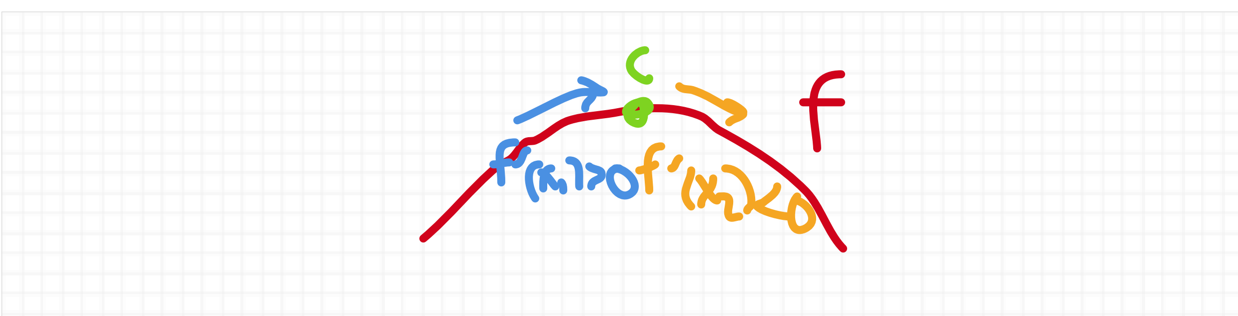
which implies that f is concave up on $(-\infty, 1)$, concave down on $(-1, 0)$, and concave up on $(0, +\infty)$.

First and Second Derivative Tests

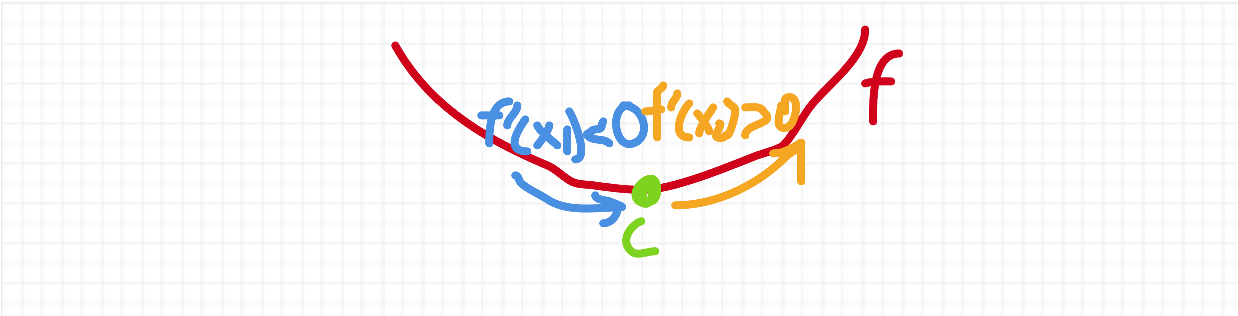
A natural question to ask is how we know a critical point is a local maximum or local minimum. We know how to find an absolute maximum/minimum in a closed interval, but we haven't thus far shown a way to figure out whether a critical point is a local maximum or minimum. We have two derivative tests for this: the first derivative test and second derivative test:

Theorem 1. *(First Derivative Test)* Assume f is a differentiable function and suppose c is a critical point.

1. If $f'(x_1) > 0$ for some $x_1 < c$ with no other critical point between it, and $f'(x_2) < 0$ for some $x_2 > c$ with no other critical point between it, then c is a local maximum



2. If $f'(x_1) < 0$ for some $x_1 < c$ with no other critical point between it, and $f'(x_2) > 0$ for some $x_2 > c$ with no other critical point between it, then c is a local minimum



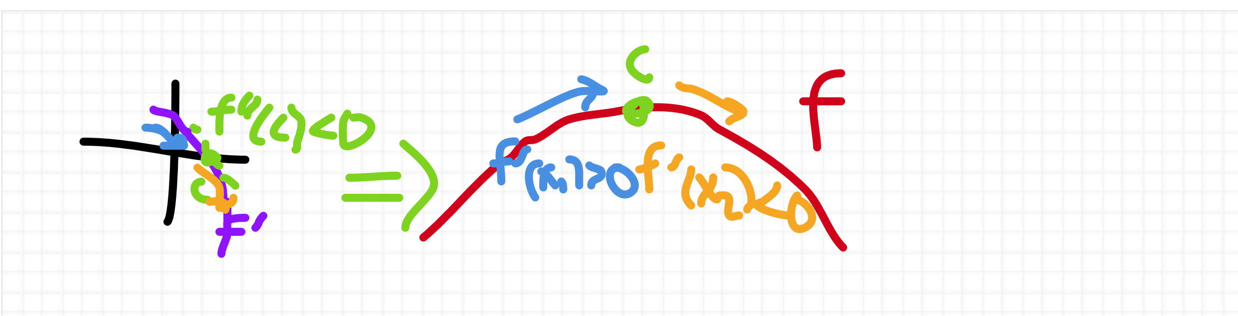
Proof.

1. It follows by **Theorem 3** in the **Increasing/Decreasing Conditions on Functions** section that f is increasing as $x \to c^-$ and decreasing as $x \to c^+$ and hence we have $f(x) \leq f(c)$ as $x \to c$.

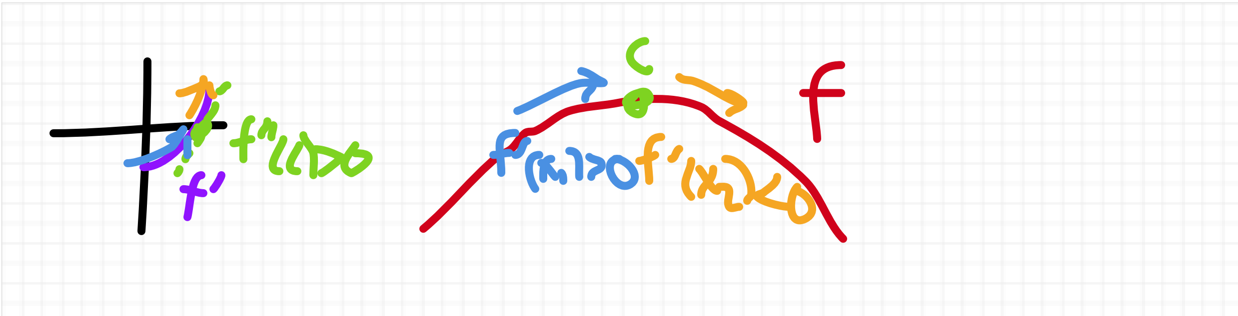
2. It follows by **Theorem 3** in the **Increasing/Decreasing Conditions on Functions** section that f is decreasing as $x \to c^-$ and increasing as $x \to c^+$ and hence we have $f(x) \geq f(c)$ as $x \to c$. $\square$

Theorem 2. *(Second Derivative Test)* Assume f is a twice differentiable function and suppose c is a critcial point.

1. If $f''(c) < 0$, then f is a local maximum



2. If $f''(c) > 0$, then f is a local mininimum



3. If $f''(c) = 0$, then the test is inconclusive

Proof.

1. By **Theorem 1** of the **Concavity and Inflection Points** section, we find $f''(c) < 0 \implies f'$ is decreasing on $c \implies f'(x) > 0$ as $x \to c^-$ and $f'(x) < 0$ as $x \to c^+$, and we conclude from the *First Derivative Test* (i.e. **Theorem 1** of this section) that c is a local maximum.

2. By **Theorem 1** of the **Concavity and Inflection Points** section, we find $f''(c) > 0 \implies f'$ is increasing on $c \implies f'(x) < 0$ as $x \to c^-$ and $f'(x) > 0$ as $x \to c^+$, and we conclude from the *First Derivative Test* (i.e. **Theorem 1** of this section) that c is a local maximum.

3. See Example 3 to see instances where f could be either an local maximum, local minimum, or neither. □

Example 1. *(from Example 5 on page 207)* We shall find the critical points of $f(x) = 2x^3 + 3x^2 - 12x$ using the first and second derivative tests.

To find the critical points, we find f' and set it equal to zero as follows

$0 = f'(x) = 6x^2 + 6x - 12 = 6\left(x^2 + x - 2\right) = (x+2)(x-1) \implies x = -2, 1$ are critical points

Using the first derivative test, we find that

$f'(-3) = 24 \implies f'(x) > 0$ on $(-\infty, -2)$
$f'(0) = -12 \implies f'(x) < 0$ on $(-2, 1)$
$f'(2) = 24 \implies f'(x) > 0$ on $(1, +\infty)$

and we conclude from that $x = -2$ is a local maximum and $x = 1$ is a local minimum

Using the second derivative test, we find that $f''(x) = 12x + 6$, and so

$f''(-2) = -18 \Longrightarrow x = -2$ is a local maximum
$f''(1) = 18 \Longrightarrow x = 1$ is a local minimum

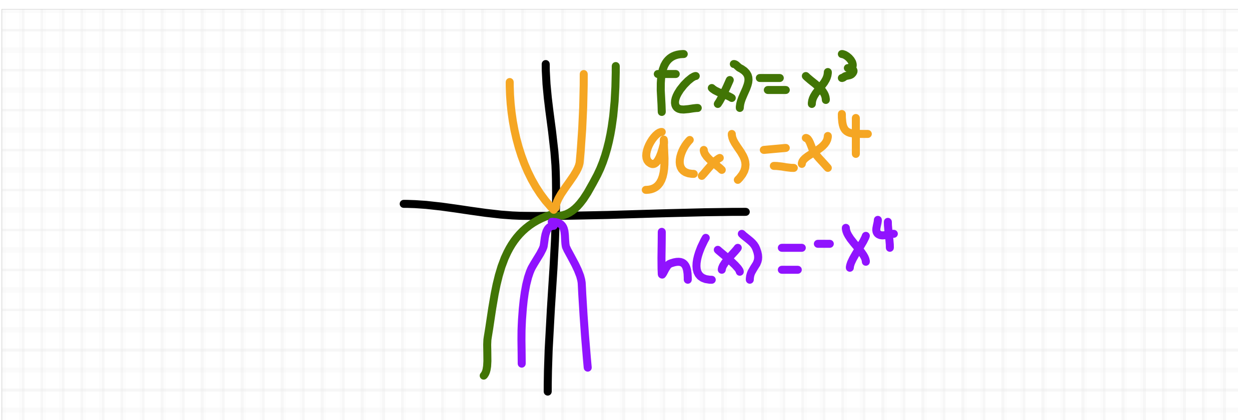
Example 2. *(Examples showing that the second derivative test is inconclusive if $f''(c) = 0$)* Take $f(x) = x^3, g(x) = x^4, h(x) = -x^4$. We find that

$f'(x) = 3x^2, g'(x) = 4x^3, h'(x) = -4x^3$

and

$f''(x) = 6x, g''(x) = 12x^2, h''(x) = -12x^2$

We can observe from the diagram below that at $x = 0$ is a critical point for $f, g,$ and h, and $f''(0) = g''(0) = h''(0) = 0$, but is a local minimum, $h(x)$ attains a local maximum (both of which can be verified using the first derivative test), but $f(x)$ is neither a local maximum or minimum.

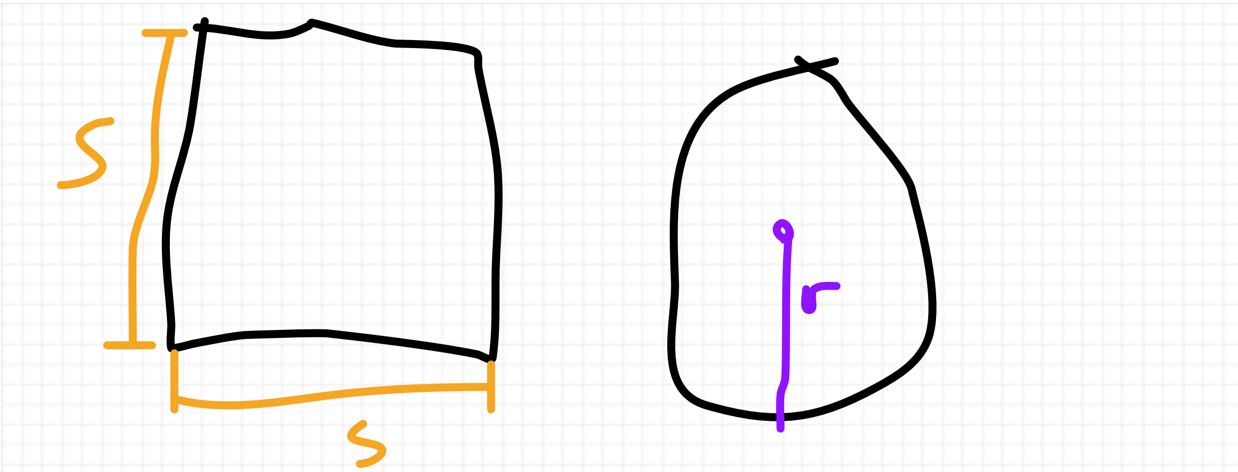


Exam 3 Review

Example 1. *(from 4.6 exercise 7 on homework 8)* A wire of length 12m is divided into two pieces and the pieces are bent into a square and a circle. How should this be done in order to minimize the sum of their areas?

Step 1. Identify the variables.

The sum of the areas of a square and a circle



We are trying to optimize the area A, which is determined by:

$$A = A_{\text{square}} + A_{\text{circle}} = s^2 + \pi r^2$$

Our constraint is the sum of the perimeters P, which is at most 12 (since the wire has length 12), which are determined by the following formula:

$$P = P_{\text{square}} + P_{\text{circle}} = 4s + 2\pi r \leq 12$$

and to optimize it, we shall set $4s + 2\pi r = P = 12$, so this gives us the following formula for s as a function of r

$$\begin{array}{lcl} 4s + 2\pi r & = & 12 \\ \quad -2\pi r & & -2\pi r \\ 4s & = & 12 - 2\pi r \\ \div 4 & & \div 4 \end{array}$$

$$s(r) = \frac{12 - 2\pi r}{4} = 3 - \frac{\pi r}{2}$$

Step 2. Convert the two variable function to a single variable function using the constraint.

$$A(r) = s(r)^2 + \pi r^2 = \left(3 - \frac{\pi r}{2}\right)^2 + \pi r^2$$

Step 3. Optimize using the opimization technique for a function defined on an interval (using the three step process for that)

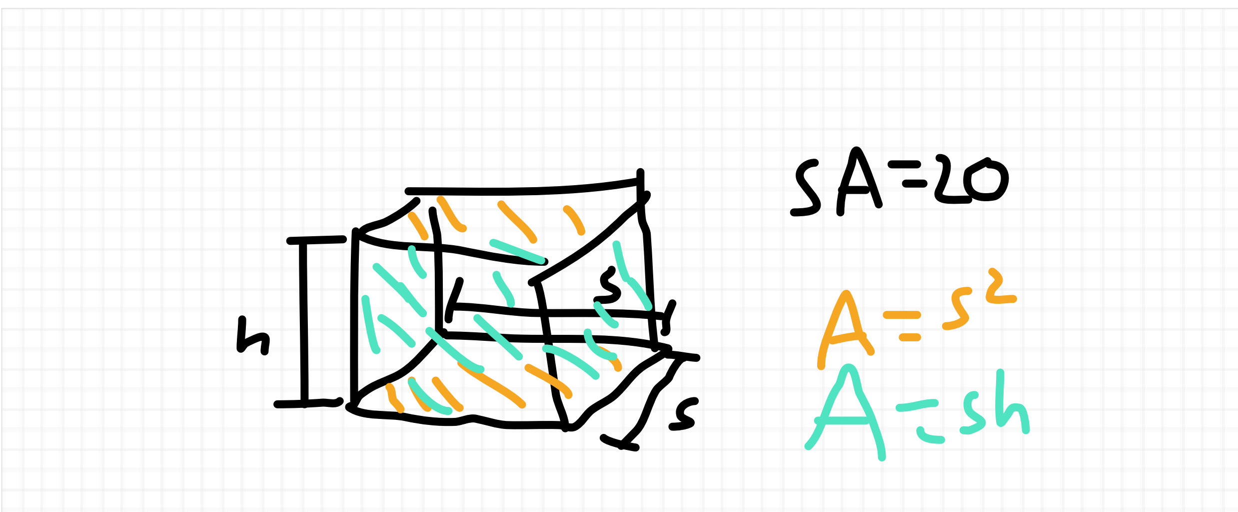
Note that by the constraint, we have

$0 \leq r \leq 12$

so we are optimizing $A(r)$ over the interval $[0, 12]$.

Find the ciritical point(s) (when $A'(r) = 0$), and plug in those plus the two endpoints ($r = 0, 12$), and the highest value is the maximum.

Example 2. *(from 4.6 exercise 13(b) on homework 8)* find the dimensions of the box with a square base with surface area 20 and maximal volume



Note that the surface area is the sum of the areas of the rectangles, so we have

$SA = 2s^2 + 4sh$

and for the volume we have

$V = s^2h$

So now we want to set this equal to a function and optimize, which we can do by taking the equation $20 = SA = 2s^2 + 4sh$ and solving for one variable (h is easier) in terms of s, so we have

$$
\begin{aligned}
2s^2 + 4sh &= 20 \\
-2s^2 \qquad &\;\; - 2s^2 \\
4sh &= 20 - 2s^2 \\
\div 4s \quad &\;\; \div 4s \\
h &= \frac{20 - 2s^2}{4s} = \frac{10 - s^2}{2s}
\end{aligned}
$$

and that leaves us with the formula of

$V(s) = s^2\left(\dfrac{10 - s^2}{2s}\right) = \dfrac{s}{2}\left(10 - s^2\right)$ over the following interval:

Note that as $h \to 0$ that means $4sh \to 0$, and the highest that s can approach is $\sqrt{10}$, since we need $2s^2 \to 20$ (which is pricisely when $s \to \sqrt{10}$ since $2\left(\sqrt{10}\right)^2 = 2 \cdot 10 = 20$) as $4sh \to 0$, so s is within the bounds of $0 \leq s \leq \sqrt{10}$

and so we're optimizing $V(s) = \dfrac{s}{2}\left(10 - s^2\right)$ over $\left[0, \sqrt{10}\right]$. (through the usual way of finding critical points, plugging in those with the endpoints, and finding the highest value)